ECE303 Receivers, Antennas, and Signals

Electronics and Communication Engineering

University of Kufa

2019

Half-Wavelength Dipole

$(l=\lambda/2)$

Half-Wave Dipole

One of the most commonly used antennas.

The arms are $\frac{\lambda}{4}$ in length and are fed at the center.

Radiation Resistance is excellent for transmission line connections:

$$\boxed{\begin{aligned} R_r &= 73 \\ Z_{in} &= 73 + j42.5 \end{aligned}}$$

To get rid of reactance, it is common practice to cut the length until it vanishes.

Directivity is also good for omnidirectional terrestrial communications

$$\boxed{D_0 = \frac{4\pi U_{max}}{P_{rad}} = 1.643 = 2.156 \text{ dBi}}$$

$\ell = \lambda / 2$

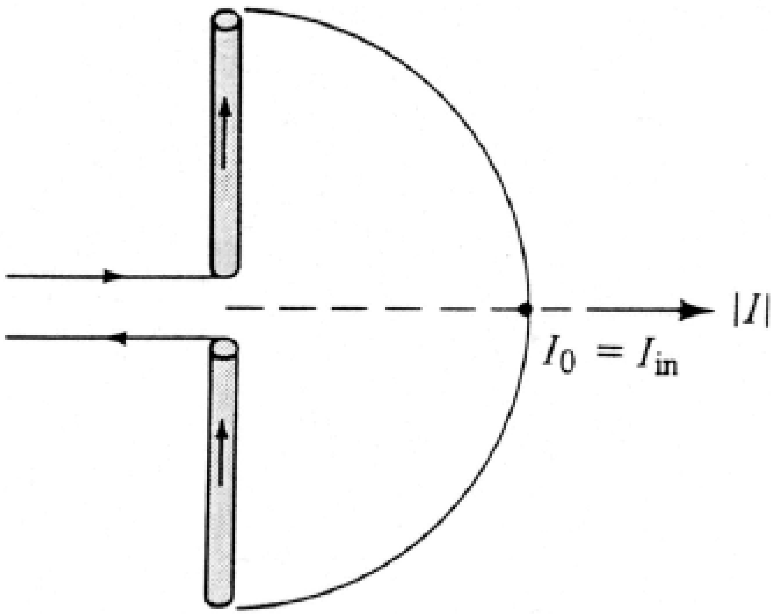


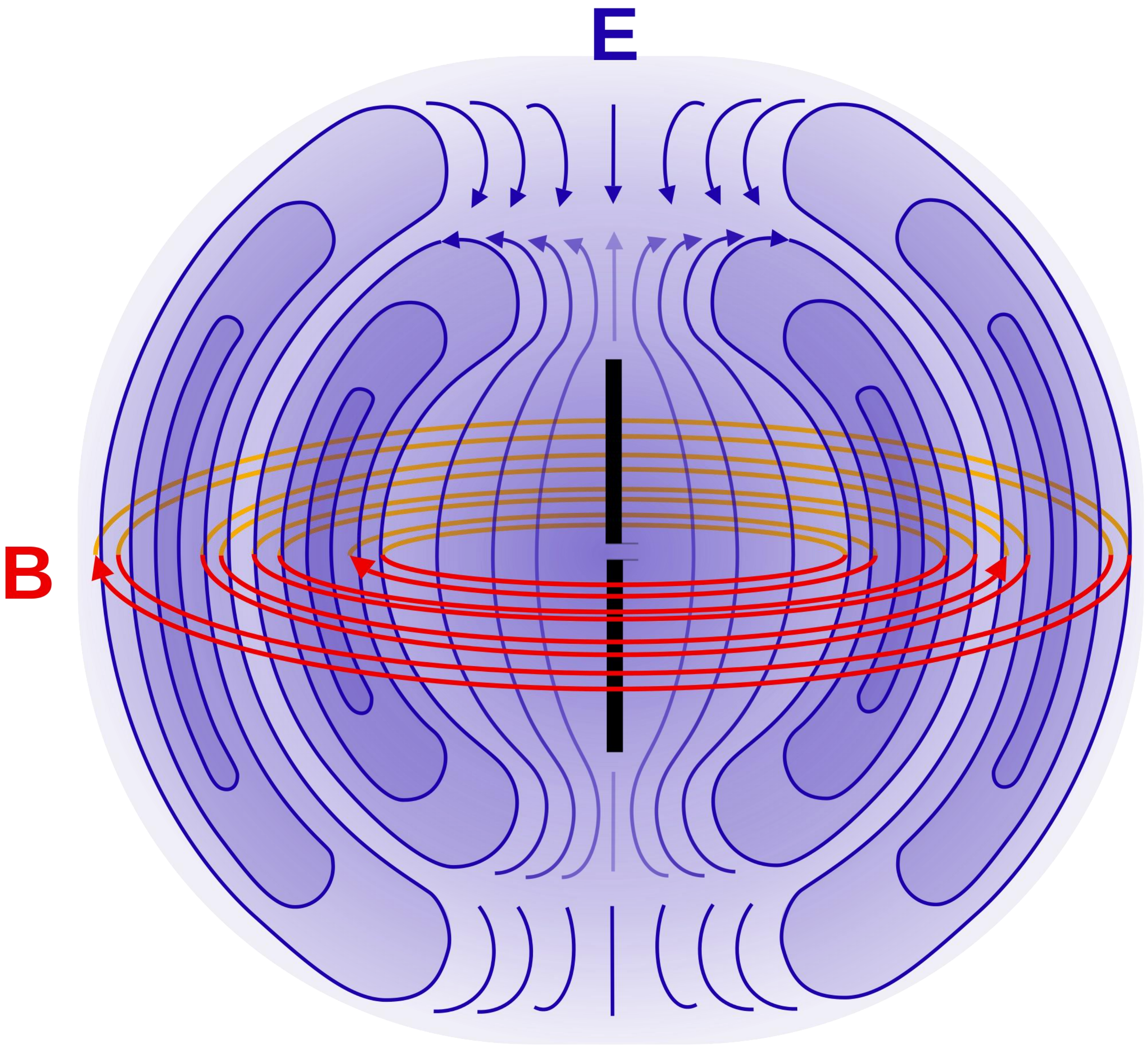
Fig. 1.16b

Chapter 1
Antennas

Half-Wavelength Dipole ($l = \lambda/2$)

$$E_\theta \simeq j\eta \frac{I_0 e^{-jkr}}{2\pi r}\left[\underbrace{\frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}}_{\text{Field Pattern}}\right] \qquad (4\text{-}84)$$

$$H_\phi \simeq j \frac{I_0 e^{-jkr}}{2\pi r}\left[\underbrace{\frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta}}_{\text{Field Pattern}}\right] = \frac{E_\theta}{\eta} \qquad (4\text{-}85)$$


E
B

$$W_{av} = \eta \frac{|I_0|^2}{8\pi^2 r^2} \underbrace{\left[\frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta} \right]^2}_{\text{Power Pattern}} \simeq \eta \frac{|I_0|^2}{8\pi^2 r^2} \sin^3\theta \quad (4\text{-}86)$$

$$U = r^2 W_{av} = \eta \frac{|I_0|^2}{8\pi^2} \underbrace{\left[\frac{\cos\left(\frac{\pi}{2}\cos\theta\right)}{\sin\theta} \right]^2}_{\text{Power Pattern}} \simeq \eta \frac{|I_0|^2}{8\pi^2} \sin^3\theta \quad (4\text{-}87)$$

Input Resistance

To calculate Input Resistance at the terminals, assume lossless antenna (no R_L) and equate the power at the input to the power at the current maximum

$$\frac{|I_{in}|^2}{2} R_{in} = \frac{|I_0|^2}{2} R_r$$

$$R_{in} = \left[\frac{I_0}{I_{in}}\right]^2 R_r$$

$$R_{in} = \text{Radiation Resistance at Input terminals}$$
$$R_r = \text{Radiation Resistance at Current Maximum}$$
$$I_0 = \text{Current Maximum}$$
$$I_{in} = \text{Current at input terminals}$$

Assuming a sinusoidal current,

$$\boxed{R_{in} = \frac{R_r}{\sin^2\left(\frac{kl}{2}\right)}}$$

Half-Wavelength Dipole

- The input impedance of the half-wavelength dipole may be obtained as

$$Z_{in} = 73.1 + j42.5\,\Omega$$

- To eliminate the imaginary part of the input impedance, we can reduce the antenna length slightly. Depending on the radius of the wire, "resonance" occurs when the length is about 0.47λ to 0.48λ.

Three-Dimensional Pattern of λ/2 Dipole

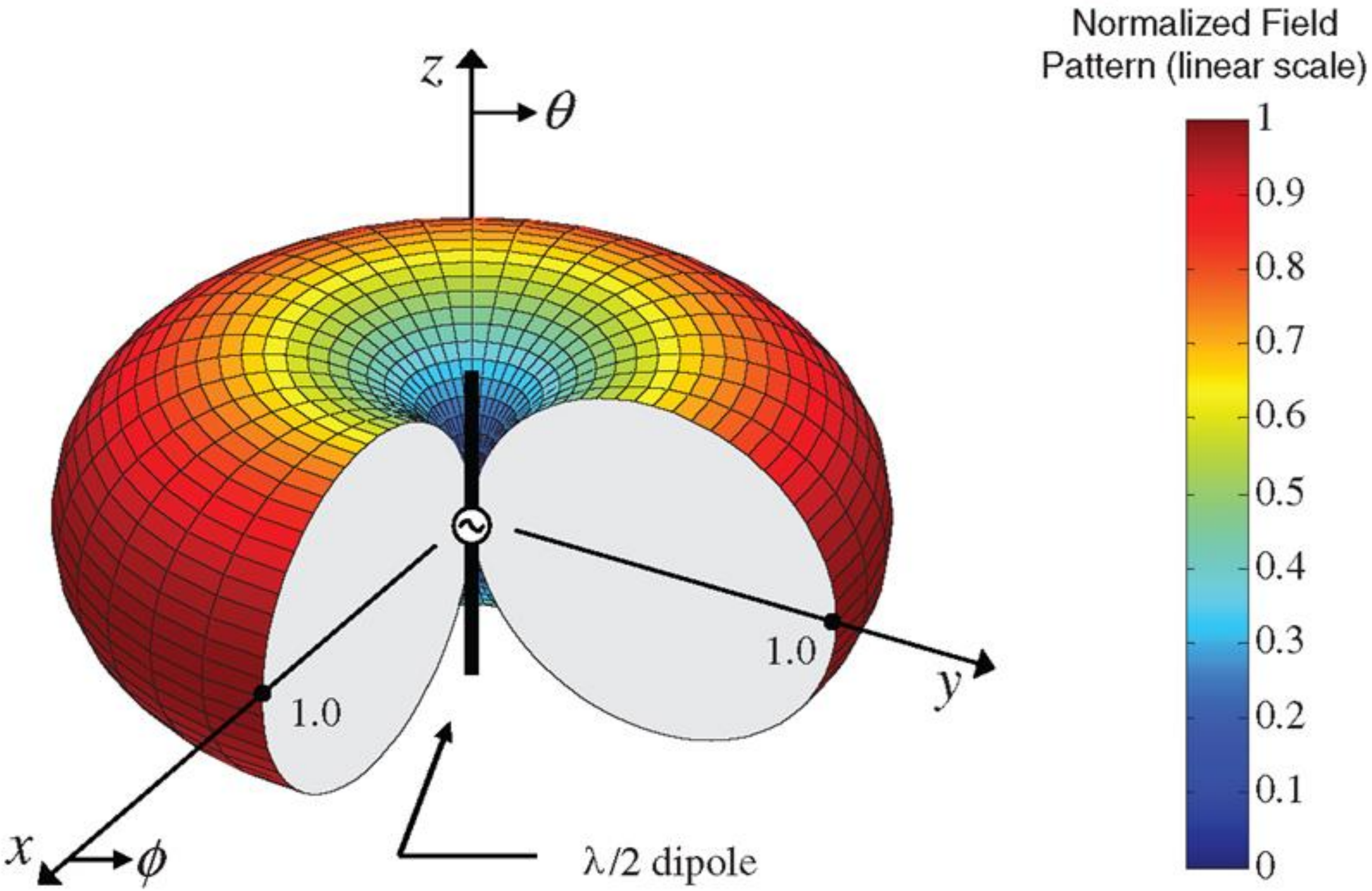


Fig. 4.12(a)

Half-Wavelength Dipole Approximations

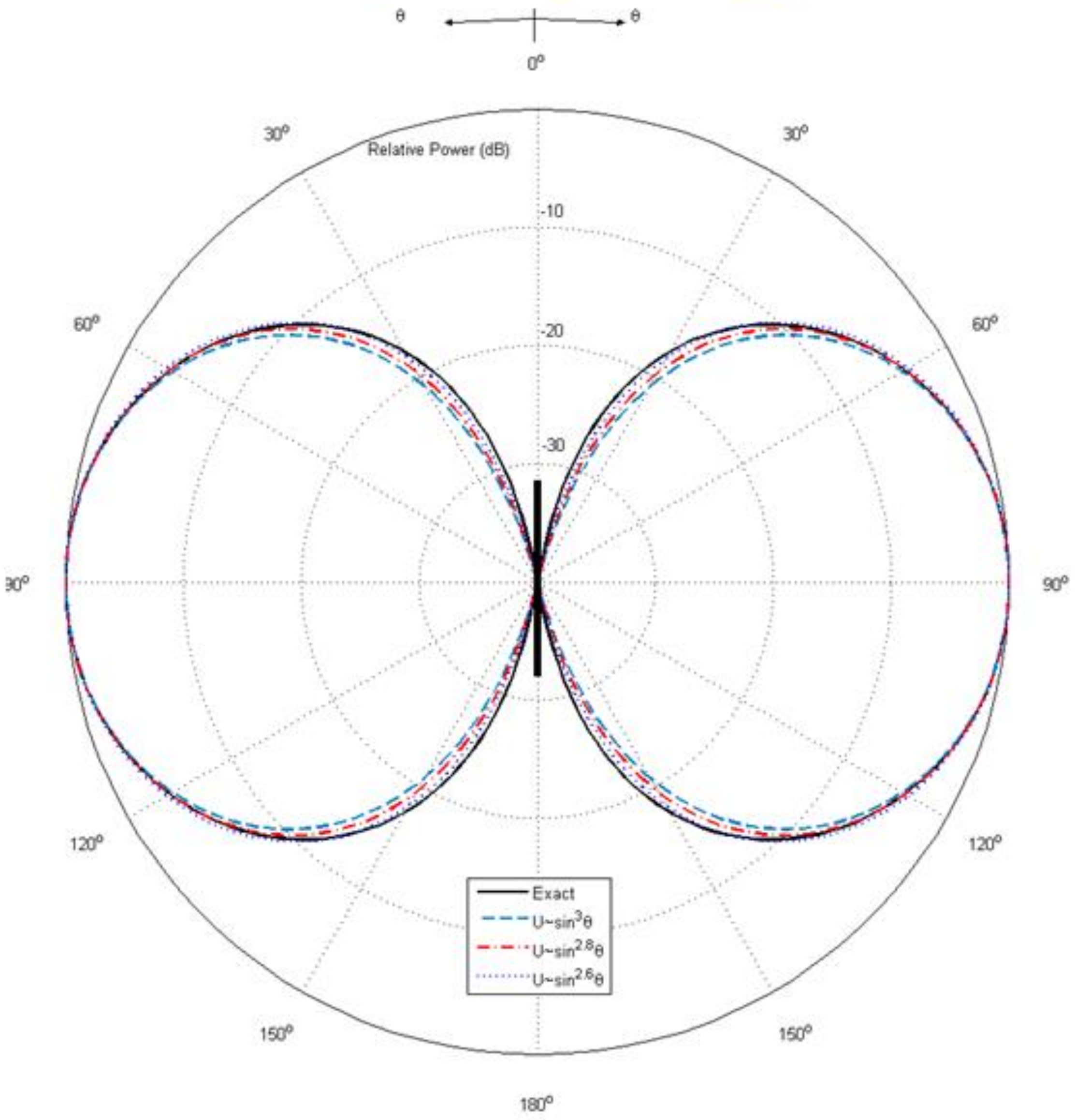


Fig. 4.12(b)

Chapter 4
Linear Wire Antennas

Dipole - Examples

A center-fed dipole of length l is attached to a balanced lossless transmission line whose characteristic impedance is $50\ \Omega$. Assuming the dipole is resonant at the given length, find the input VSWR when

(a) $l = \frac{\lambda}{4}$ (b) $l = \frac{\lambda}{2}$ (c) $l = \frac{3\lambda}{4}$ (d) $l = \lambda$

$$VSWR = \frac{1 + |\Gamma|}{1 - |\Gamma|} \qquad \Gamma = \frac{R_{in} - Z_0}{R_{in} + Z_0} \qquad R_{in} = \frac{R_r}{\sin^2\left(\frac{kl}{2}\right)}$$

$$R_r\Big|_{l=\frac{\lambda}{4}} \cong 6.84$$

$$R_r\Big|_{l=\frac{\lambda}{2}} \cong 73$$

$$R_r\Big|_{l=\frac{3\lambda}{4}} \cong 372$$

$$R_r\Big|_{l=\frac{\lambda}{4}} = \infty$$